Started on	2024-10-15 21:37
State	Finished
Completed on	2024-10-15 23:58
Time taken	2 hours 21 mins
Marks	730.00/919.00
Grade	**7.94** out of 10.00 (**79**%)

Question **1**
Correct
Mark 10.00 out of 10.00

In segmentation, each linear address is specified by :

Select one:
1. a key
2. a segment value
3. a segment base address and offset ✓
4. a segment selector

The correct answer is: a segment base address and offset

Question **2**
Incorrect
Mark 0.00 out of 5.00

The segment base address contains the :

Select one:
1. segment offset
2. segment length
3. starting linear address of the segment of the process
4. starting physical address of the segment in memory ✗

The correct answer is: starting linear address of the segment of the process

Question **3**
Correct
Mark 5.00 out of 5.00

The address of a page directory table in memory is pointed by ……………………..

Select one:
1. page pointer
2. stack pointer
3. control register ✓
4. program counter

The correct answer is: control register

Question 4

Correct

Mark 5.00 out of 5.00

The physical address of the Page Directory Table in main memory is stored in ··········· processor control register.

Select one:

1. cr0
2. cr4
3. cr3
4. cr2

The correct answer is: cr3

Question 5

Correct

Mark 5.00 out of 5.00

Linux's kernel memory allocator uses Slab Allocator algorithm on top of a ···············.···. approach.

Select one:

1. Buddy system
2. Power-of-two free lists
3. McKusick-Karels allocator
4. Dynix allocator

The correct answer is: Buddy system

Question 6

Correct

Mark 5.00 out of 5.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x03000000, Limit = 0x3FFFF, G=0

Compute the size or length of the Linear Segment in KB

Answer: 256

The correct answer is: 256KB

Question 7

Correct

Mark 5.00 out of 5.00

If any parent process terminates without reaping a child, then the orphaned child will be reaped by ···············.

Select one:

1. init process
2. process scheduler
3. Zombie process
4. swapper process

The correct answer is: init process

Question 8

Correct

Mark 5.00 out of 5.00

Consider two-level Paging unit using a 32-bit virtual address divided into three fields -Page directory containing most significant 10 bits, Page Table containing intermediate 10 bits and page offset containing least significant 12 bits.

Determine the number of entries in each Page Table.

Select one:

1. 512
2. 2048
3. 1024 ✓
4. 4096

The correct answer is: 1024

Question 9

Correct

Mark 5.00 out of 5.00

Assume that the kernel has assigned the linear address space between 0x30000000 and 0x30FFFFFF to a running process. Two-level Paging is used with page size 4K.

Determine the number of Page Directories required.

Select one:

1. 1024
2. 4
3. 2
4. 1 ✓

The correct answer is: 1

Question 10

Correct

Mark 10.00 out of 10.00

TLB stands for ………………………………………………………….

Select one:

1. Translation Look-alike Buffer
2. Translation Look-aside Buffer ✓
3. Transformation Look-alike Buffer
4. Transformation Look-aside Buffer

The correct answer is: Translation Look-aside Buffer

Question 11
Incorrect
Mark 0.00 out of 5.00

To speed up the translation of logical addresses into linear addresses, the Intel processor provides ………………………………………………. Registers.

Select one:
1. Offset Registers
2. Programmable Registers
3. Segmentation Registers ✗
4. Non-Programmable Registers

The correct answer is: Non-Programmable Registers

Question 12
Correct
Mark 5.00 out of 5.00

If a child process terminates before the parent process dies then the child process is termed as ……………………..

Select one:
1. Bad process
2. Orphan process
3. Dead process
4. Zombie process ✓

The correct answer is: Zombie process

Question 13
Correct
Mark 5.00 out of 5.00

Operating System maintains the page table for …………………..

Select one:
1. each address
2. each thread
3. each process ✓
4. each instruction

The correct answer is: each process

Question 14
Correct
Mark 5.00 out of 5.00

Consider two-level Paging unit using a 32-bit virtual address divided into three fields -Page directory containing most significant 10 bits, Page Table containing intermediate 10 bits and page offset containing least significant 12 bits.

Determine the number of entries in Page directory

Select one:
1. 1024 ✓
2. 2048
3. 512
4. 4096

The correct answer is: 1024

Question 15
Incorrect
Mark 0.00 out of 10.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x03000000, Limit = 0x3FFF, G=1

Determine the ending address of the Linear Segment

Answer: 0x03003FFF ✗

The correct answer is: 0x06FFFFFF

Question 16
Correct
Mark 3.00 out of 3.00

Is the following statement TRUE or FALSE ……………

"Each process descriptor includes its own TSS segment and a pointer to its LDT segment, which is also located inside the kernel data segment"

Select one:

- (●) True ✓
- ( ) False

The correct answer is 'True'.

Question 17
Correct
Mark 15.00 out of 15.00

Given gdtr = 0x0B000000 pointing to the GDT containing Segment Descriptors.

If the logical address generated by the processor is given as follows :

CS register contains the value 0x63 and IP register has the value 0x500 then

Determine the starting address of the corresponding Segment Descriptor stored in the GDT (i. e, address of the corresponding GDT entry) in hex.

Answer: 0x0B000060 ✓

The correct answer is: 0x0B000060

Question 18
Correct
Mark 5.00 out of 5.00

Consider four-level Paging unit using a 48-bit virtual address divided into five fields - Global Page directory containing most significant 9 bits, Upper Page directory containing next 9 bits, Middle Page directory containing next 9 bits, Page Table containing next 9 bits and page offset containing least significant 12 bits.

Determine the maximum number of Upper Page Directory tables.

Select one:

- ( ) 1. 256
- ( ) 2. 2048
- (●) 3. 512 ✓
- ( ) 4. 1024

The correct answer is: 512

Question 19
Correct
Mark 5.00 out of 5.00

Number of Non-Programmable Registers inside the processor to hold segment desriptors is ....

Select one:
- 1. 6 ✓
- 2. 64
- 3. 8191
- 4. 32

The correct answer is: 6

Question 20
Incorrect
Mark 0.00 out of 5.00

How many privilege levels are in Linux?

Select one:
- 1. 4 ✗
- 2. 2
- 3. 3
- 4. 1

The correct answer is: 2

Question 21
Incorrect
Mark 0.00 out of 5.00

To reap a child process, ... system call allows a parent process to wait until one of its children terminates.

Answer: wait() ✗

The correct answer is: wait( )

Question 22
Correct
Mark 10.00 out of 10.00

Consider four-level Paging unit using a 48-bit virtual address divided into five fields - Global Page directory containing most significant 9 bits, Upper Page directory containing next 9 bits, Middle Page directory containing next 9 bits, Page Table containing next 9 bits and page offset containing least significant 12 bits.

Determine the number of entries in Global Page directory Table/Upper Page Directory Table/Middle Page Directory Table/Page Table.

Select one:
- 1. 512 ✓
- 2. 256
- 3. 1024
- 4. 2048

The correct answer is: 512

Question 23
Correct
Mark 5.00 out of 5.00

Functions that can modify only local variables and do not alter global data structures are called as……………………

Select one:
- 1. System call
- 2. Kernel Control path
- 3. Signal
- 4. Re-entrant ✓

The correct answer is: Re-entrant

Question 24
Correct
Mark 5.00 out of 5.00

…….. Loads a new program into an existing content and then executes it.

Select one:
- 1. New
- 2. Execve ✓
- 3. Fork
- 4. Init

The correct answer is: Execve

Question 25
Incorrect
Mark 0.00 out of 10.00

Assume that the kernel has assigned the linear address space between 0x30000000 and 0x30FFFFFF to a running process. Two-level Paging is used with page size 4K.

Suppose that the process needs to read the byte at linear address 0x30F42646 then determine which entry or byte of the Page frame in the memory is accessed.

Answer: 0x046 ✗

The correct answer is: 0x646

Question 26
Correct
Mark 5.00 out of 5.00

Control flow passes from one process to another process via …………………..

Select one:
- 1. Context switching ✓
- 2. Call instruction
- 3. Return instruction
- 4. Jump instruction

The correct answer is: Context switching

Question 27

Incorrect

Mark 0.00 out of 3.00

A .... is very similar to a semaphore, but it has no process list: when a process finds the lock closed by another process, it revolves around it repeatedly, executing a tight instruction loop until the lock becomes open.

Answer: spinlock ✗

The correct answer is: spin lock

Question 28

Correct

Mark 5.00 out of 5.00

System call to terminate a process is ....

Answer: exit() ✓

The correct answer is: exit()

Question 29

Correct

Mark 10.00 out of 10.00

Assume that the kernel has assigned the linear address space segment between 0x40000000 and 0x407FFFFF to a running process.

Two-level Paging is used with page size 4K. Determine the size of the segment allocated to the process.

Select one:

1. 32MB
2. 8MB ✓
3. 64MB
4. 4MB

The correct answer is: 8MB

Question 30

Correct

Mark 10.00 out of 10.00

In 80x86 processors, Logical address associated with code is specified using the registers ....

Select one:

1. SS, SP
2. CS, DI
3. DS, SI
4. CS, IP ✓

The correct answer is: CS, IP

Question 31
Correct
Mark 2.00 out of 2.00

Linux's kernel memory allocator uses Slab Allocator algorithm on top of a ……………..…. approach.

Select one:
1. Dynix allocator
2. McKusick-Karels allocator
3. Power-of-two free lists
4. Buddy system ✓

The correct answer is: Buddy system

Question 32
Correct
Mark 5.00 out of 5.00

The physical address of the Page Directory Table is stored in ………… processor control register.

Select one:
1. cr3 ✓
2. cr2
3. cr0
4. cr1

The correct answer is: cr3

Question 33
Incorrect
Mark 0.00 out of 2.00

State whether the following statement is TRUE or FALSE -

"The fourth gigabyte of linear addresses is reserved for the kernel, while the first three gigabytes are accessible from both the kernel and the user programs."

Select one:
True
False ✗

The correct answer is 'True'.

Question 34
Correct
Mark 5.00 out of 5.00

If a child process terminates after the parent process dies then the child process is termed as …………………..

Select one:
1. Zombie process
2. Bad process
3. Orphan process ✓
4. Dead process

The correct answer is: Orphan process

Question 35

Correct

Mark 15.00 out of 15.00

Consider four-level Paging unit using a 48-bit virtual address divided into five fields - Global Page directory containing most significant 9 bits, Upper Page directory containing next 9 bits, Middle Page directory containing next 9 bits, Page Table containing next 9 bits and page offset containing least significant 12 bits.

Determine the maximum number of pages that can be stored and total amount of memory required to store these pages.

Select one:

1. 256x256x256x256, 16TB
2. 2048x2048x2048x2048, 64PB
3. 1024x1024x1024x1024, 4PB
4. 512x512x512x512, 256 TB ✓

The correct answer is: 512x512x512x512, 256 TB

Question 36

Correct

Mark 10.00 out of 10.00

Assume that the kernel has assigned the linear address space between 0x30000000 and 0x30FFFFFF to a running process. Two-level Paging is used with page size 4K.

Compute the size or length of the Linear Segment

Select one:

1. 128MB
2. 4MB
3. 16MB ✓
4. 32MB

The correct answer is: 16MB

Question 37

Incorrect

Mark 0.00 out of 5.00

The system call used by a process to acquire System V IPC Semaphore resource is …………………….

Answer: semop() ✗

The correct answer is: semget( )

Question 38
Correct
Mark 10.00 out of 10.00

Assume that the kernel has assigned the linear address space between 0x30000000 and 0x30FFFFFF to a running process. Two-level Paging is used with page size 4K.

Determine the number of Page tables required

Select one:
1. 2
2. 1
3. 1024
4. 4

The correct answer is: 4

Question 39
Correct
Mark 10.00 out of 10.00

In the Linux Physical Memory Layout, PAGE_OFFSET macro refers to the address ………………….which indicates the starting address of the fourth gigabyte of linear addresses reserved for the kernel(Kernel Memory)

Select one:
1. 0xFFFFFFFF
2. 0x00100000
3. 0x90000000
4. 0xC0000000

The correct answer is: 0xC0000000

Question 40
Incorrect
Mark 0.00 out of 5.00

……………………….. translates logical addresses to linear addresses

Select one:
1. Paging Unit
2. Segmentation Unit
3. Memory Controller unit
4. Arithmetic Logic Unit

The correct answer is: Segmentation Unit

Question 41
Correct
Mark 5.00 out of 5.00

Physical memory is broken into fixed-sized blocks called ______________.

Select one:
1. Page frames
2. Segments
3. Directories
4. Pages

The correct answer is: Page frames

Question 42
Correct
Mark 10.00 out of 10.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0xA0000000, Limit = 0xFFFFF, G=0

Determine the ending address of the Linear Segment in hex.

Answer: 0xA00FFFFF

The correct answer is: 0xA00FFFFF

Question 43
Correct
Mark 10.00 out of 10.00

Given a Virtual Memory System having 48-bit virtual address, 4KB Page size and using a single Page table (i.e., just one Page directory - one level mapping) with 8 byte PTE (Page table entry),

Determine the amount of Main Memory required to store a single Page Table(i.e., just one Page Directory).

Select one:
1. 256GB
2. 64GB
3. 512GB
4. 128GB

The correct answer is: 512GB

Question 44
Correct
Mark 10.00 out of 10.00

Assume that the kernel has assigned the linear address space between 0x30000000 and 0x30FFFFFF to a running process. Two-level Paging is used with page size 4K.

Compute the number of pages in the Linear Segment

Select one:
1. 2048
2. 1024
3. 8192
4. 4096

The correct answer is: 4096

Question 45
Correct
Mark 5.00 out of 5.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x03000000, Limit = 0x3FFF, G=0

Compute the size or length of the Linear Segment in KB

Answer: 16

The correct answer is: 16KB

Question 46

Correct

Mark 10.00 out of 10.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x00000000, Limit = 0xFFFFF, G=1

Compute the size or length of the Linear Segment in GB.

Answer: 4

The correct answer is: 4GB

Question 47

Correct

Mark 2.00 out of 2.00

The system call used to create a light-weight process or thread is ……………. …

Select one:

1. vfork
2. fork
3. flock
4. Clone

The correct answer is: Clone

Question 48

Correct

Mark 10.00 out of 10.00

Given a Virtual Memory System having 32-bit virtual address, 4KB Page size and using a single Page table (i.e., just one Page directory - one level mapping) with 4 byte PTE (Page table entry),

Determine the amount of Main Memory required to store a single Page Table(i.e., just one Page Directory).

Select one:

1. 4GB
2. 2MB
3. 4MB
4. 2GB

The correct answer is: 4MB

Question 49
Correct
Mark 5.00 out of 5.00

Current Privilege Level of the CPU reflects whether the processor is in User or Kernel Mode and is specified by the ……. field of the Segment Selector stored in the CS register.

Select one:
- 1. INDEX
- 2. TI
- 3. RPL ✓
- 4. DPL

The correct answer is: RPL

Question 50
Correct
Mark 5.00 out of 5.00

Indicate which one of these is a Non-System Segment ....

Select one:
- 1. Local Descriptor Table Segment
- 2. Code Segmentation Register
- 3. Task State Segment
- 4. Data Segment ✓

The correct answer is: Data Segment

Question 51
Correct
Mark 5.00 out of 5.00

……………………. translates linear addresses to physical addresses.

Select one:
- 1. Segmentation Unit
- 2. Paging Unit ✓
- 3. Arithmetic Logic Unit
- 4. Memory Controller unit

The correct answer is: Paging Unit

Question 52
Correct
Mark 10.00 out of 10.00

Assume that the kernel has assigned the linear address space between 0x30000000 and 0x30FFFFFF to a running process. Two-level Paging is used with page size 4K.

Determine how many entries in the Page Directory are used.

Select one:
- 1. 4 ✓
- 2. 1024
- 3. 1
- 4. 8

The correct answer is: 4

Question 53
Correct
Mark 10.00 out of 10.00

PAE stands for ....

Answer: Physical Address Extension ✓

The correct answer is: Physical Address Extension

Question 54
Correct
Mark 5.00 out of 5.00

Physical memory is broken into fixed-sized blocks called _____________

Select one:
- 1. Directories
- 2. Page Frames ✓
- 3. Segments
- 4. Pages

The correct answer is: Page Frames

Question 55
Correct
Mark 5.00 out of 5.00

The kernel interacts with I/O devices by means of ....

Answer: device drivers ✓

The correct answer is: device drivers

Question 56
Correct
Mark 10.00 out of 10.00

In Intel processors, Paging is enabled by setting the .... bit in ...... register

Select one:
- 1. PG, CR0 ✓
- 2. PG, CR4
- 3. PSE, CR4
- 4. PSE, CR0

The correct answer is: PG, CR0

Question 57
Correct
Mark 10.00 out of 10.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x00000000, Limit = 0xFFFFF, G=1

Determine the ending address of the Linear Segment in hex.

Answer: 0xFFFFFFFF ✓

The correct answer is: 0xFFFFFFFF

Question 58
Correct
Mark 3.00 out of 3.00

Processes can also share parts of their address space as a kind of interprocess communication, using the ……………… technique introduced in System V and supported by Linux.

Answer: shared memory

The correct answer is: shared memory

Question 59
Incorrect
Mark 0.00 out of 5.00

Select the synchronization mechanism which is most suitable for Uniprocessor systems

Select one:
1. Interrupt disabling
2. Kernel preemption disabling (Non-Preemptive Kernels)
3. Semaphores
4. Spin locks

The correct answer is: Semaphores

Question 60
Correct
Mark 5.00 out of 5.00

Run time mapping from linear to physical address is done by ……………………..

Select one:
1. I/O
2. APIC
3. MMU
4. CPU

The correct answer is: MMU

Question 61
Correct
Mark 10.00 out of 10.00

Consider four-level Paging unit using a 48-bit virtual address divided into five fields - Global Page directory containing most significant 9 bits, Upper Page directory containing next 9 bits, Middle Page directory containing next 9 bits, Page Table containing next 9 bits and page offset containing least significant 12 bits.

Determine the maximum number of Middle Page Directory tables

Select one:
1. 1024x1024
2. 512x512
3. 2048x2048
4. 256x256

The correct answer is: 512x512

Question 62
Correct
Mark 3.00 out of 3.00

Linux supports the ………… system call, which allows part of a file or the memory residing on a device to be mapped into a part of a process address space.

Answer: mmap()

The correct answer is: mmap()

Question 63
Correct
Mark 10.00 out of 10.00

Consider two-level Paging unit using a 32-bit virtual address divided into three fields -Page directory containing most significant 10 bits, Page Table containing intermediate 10 bits and page offset containing least significant 12 bits. Determine the maximum number of Page tables that can be stored.

Select one:
- 1. 1024
- 2. 2048
- 3. 4096
- 4. 512

The correct answer is: 1024

Question 64
Correct
Mark 5.00 out of 5.00

Assume that the kernel has assigned the linear address space between 0x30000000 and 0x30FFFFFF to a running process. Two-level Paging is used with page size 4K.

Determine how many entries are used in each Page table.

Select one:
- 1. 8
- 2. 4
- 3. 1024
- 4. 1

The correct answer is: 1024

Question 65
Correct
Mark 5.00 out of 5.00

…………………….is a sequence of instructions executed by the kernel to handle a system call, an exception or an interrupt.

Select one:
- 1. Re-entrant
- 2. Kernel Control path
- 3. Signal
- 4. System call

The correct answer is: Kernel Control path

Question 66
Correct
Mark 3.00 out of 3.00

The function to get the PID of a calling process is ……………………………………..

Select one:
1. getppgrp
2. getpid
3. getppid
4. getpgrp

The correct answer is: getpid

Question 67
Correct
Mark 5.00 out of 5.00

…………………..is a short message sent to a process by the kernel to notify the process that certain event has occurred.

Select one:
1. Signal
2. Re-entrant
3. Kernel Control path
4. System call

The correct answer is: Signal

Question 68
Correct
Mark 5.00 out of 5.00

…………. uses Copy On Write (COW) technique that allows both the parent and the child processes to read the same physical pages. Whenever either one tries to write on a physical page, the kernel copies its contents into a new physical page that is assigned to the writing process.

Select one:
1. clone() system call
2. flock() system call
3. fork() system call
4. vfork() system call

The correct answer is: fork() system call

Question 69
Correct
Mark 10.00 out of 10.00

Select synchronization mechanisms which are suitable for Multiprocessor systems.

Select one or more:

- [x] 1. Spin locks ✓
- [ ] 2. Kernel preemption disabling (Non-Preemptive Kernels)
- [x] 3. Semaphores ✓
- [ ] 4. Interrupt disabling

The correct answers are: Semaphores, Spin locks

Question 70
Correct
Mark 5.00 out of 5.00

Indicate which one of these is a System Segment ....

Select one:

- ( ) 1. Data Segment
- ( ) 2. Stack Segment
- ( ) 3. Code Segment
- (•) 4. Task State Segment ✓

The correct answer is: Task State Segment

Question 71
Incorrect
Mark 0.00 out of 15.00

In Linux, each GDT has total of……...entries out of which ………contain segment descriptors and remaining………..are null, unused, or reserved entries.

Select one:

- ( ) 1. 32, 18, 14
- ( ) 2. 32, 16, 16
- ( ) 3. 64, 32, 32
- (•) 4. 8192, 4096, 4096 ✗

The correct answer is: 32, 18, 14

Question 72
Correct
Mark 5.00 out of 5.00

The …………..flag in the page directory entry enables large page sizes 4 MB when PSE is enabled.

Answer: PS ✓

The correct answer is: PS

Question 73
Incorrect
Mark 0.00 out of 10.00

When the CPL is equal to 0, the ds register must contain the Segment Selector of the ………………….segment.

Answer: system

The correct answer is: Kernel Data

Question 74
Correct
Mark 10.00 out of 10.00

The .... system call forces disk synchronization by writing all of the "dirty" buffers (i.e., all the buffers whose contents differ from that of the corresponding disk blocks) into disk.

Answer: sync()

The correct answer is: sync()

Question 75
Incorrect
Mark 0.00 out of 10.00

.... are usually created during system startup and remain alive until the system is shut down

Answer: System processes

The correct answer is: Kernel threads

Question 76
Incorrect
Mark 0.00 out of 15.00

Given gdtr = 0x0C000000 pointing to the GDT containing Segment Descriptors.

If the logical address generated by the processor is given as follows :

CS register contains the value 0xCB and IP register has the value 0x00000000 then

Determine the starting address of the corresponding Segment Descriptor stored in the GDT (i. e, address of the corresponding GDT entry) in hex.

Answer: 0x0C000258

The correct answer is: 0x0C0000C8

Question 77
Correct
Mark 5.00 out of 5.00

The system call used by a process to acquire System V IPC Shared Memory resource is ………………….

Answer: shmget()

The correct answer is: shmget( )

Question 78
Correct
Mark 10.00 out of 10.00

In the Linux Physical Memory Layout, the symbol _text, which corresponds to physical address …………………………………… denotes the address of the first byte of kernel code.

Select one:
1. 0x00100000 ✓
2. 0x90000000
3. 0xC0000000
4. 0xFFFFFFFF

The correct answer is: 0x00100000

Question 79
Correct
Mark 5.00 out of 5.00

Current Privilege Level of the CPU reflects whether the processor is in User or Kernel Mode and is specified by the …….. field of the Segment Selector stored in the CS register.

Select one:
1. RPL ✓
2. DPL
3. TI
4. INDEX

The correct answer is: RPL

Question 80
Incorrect
Mark 0.00 out of 5.00

For the kernel to manage processes, each process is represented by a ………… that includes information about the current state of the process.

Answer: process control block ✗

The correct answer is: process descriptor

Question 81
Correct
Mark 5.00 out of 5.00

………………….. is an explicit request made by the process to the kernel to get its service via interrupt.

Select one:
1. Re-entrant
2. Signal
3. Kernel Control path
4. System call ✓

The correct answer is: System call

Question 82
Correct
Mark 5.00 out of 5.00

To avoid the race condition, the number of processes that may be simultaneously inside their critical section is..............

Select one:
1. 2
2. 3
3. 1 ✓
4. 4

The correct answer is: 1

Question 83
Correct
Mark 15.00 out of 15.00

Consider two-level Paging unit using a 32-bit virtual address divided into three fields -Page directory containing most significant 10 bits, Page Table containing intermediate 10 bits and page offset containing least significant 12 bits.

Determine the maximum amount of memory required to store all pages, Page directory & Page tables assuming that each entry in the page directory and page tables take 32-bits (each page table entry size =32-bits).

Select one:
1. 8KB + 16MB + 16GB
2. 2KB + 1MB + 1GB
3. 4KB + 4MB + 4GB ✓
4. 16KB + 64MB + 64GB

The correct answer is: 4KB + 4MB + 4GB

Question 84
Correct
Mark 5.00 out of 5.00

The system call used by a process to acquire System V IPC Message queue resource is …………………….

Answer: msgget() ✓

The correct answer is: msgget( )

Question 85
Correct
Mark 10.00 out of 10.00

In Intel processors, main memory greater than 4GB can be used by setting the ……. Flag in the ……..Register.

Select one:
1. PAE, cr4 ✓
2. PAE, cr0
3. PG, cr4
4. PG, cr0

The correct answer is: PAE, cr4

Question 86
Correct
Mark 5.00 out of 5.00

The ....register in the processor provides the base address of the Local Descriptor Table in the main memory.

Select one:
- 1. gdtr
- 2. CR0
- 3. CR3
- 4. ldtr ✓

The correct answer is: ldtr

Question 87
Correct
Mark 5.00 out of 5.00

The value 3 in 2-bit Current Privilege Level(CPL) field of CS register refers to ....

Select one:
- 1. User Mode ✓
- 2. Kernel mode
- 3. Real Mode
- 4. Protected mode

The correct answer is: User Mode

Question 88
Correct
Mark 10.00 out of 10.00

When the CPL is equal to 3, the DS register must contain the Segment Selector of the ········· segment.

Answer: user data ✓

The correct answer is: user Data

Question 89
Correct
Mark 10.00 out of 10.00

Assume that the kernel has assigned the linear address space segment between 0x40000000 and 0x407FFFFF to a running process.

Two-level Paging is used with page size 4K. Determine the number of pages allocated in the segment for the process.

Select one:
- 1. 512
- 2. 4096
- 3. 2048 ✓
- 4. 1024

The correct answer is: 2048

Question 90
Correct
Mark 2.00 out of 2.00

Control flow passes from one process to another process via …………………..

Select one:
- 1. Context switching ✓
- 2. Jump instruction
- 3. Call instruction
- 4. Return instruction

The correct answer is: Context switching

Question 91
Incorrect
Mark 0.00 out of 10.00

Full form of COW ……..

Answer: Copy-On-Write ✗

The correct answer is: Copy On Write

Question 92
Correct
Mark 5.00 out of 5.00

…………………………………is a small set-associative hardware cache inside the CPU which maps linear (virtual) page numbers to physical page numbers.

Select one:
- 1. Translation Look-aside Buffer(TLB) ✓
- 2. Split Cache L1
- 3. Memory Management Unit (MMU)
- 4. Unified Cache L2

The correct answer is: Translation Look-aside Buffer(TLB)

Question 93
Correct
Mark 10.00 out of 10.00

Assume that the kernel has assigned the linear address space segment between 0x30000000 and 0x300FFFFF to a running process.

Two-level Paging is used with page size 4K. Determine the size of the segment allocated for the process.

Select one:
- 1. 256MB
- 2. 128MB
- 3. 64MB
- 4. 1MB ✓

The correct answer is: 1MB

Question 94

Incorrect

Mark 0.00 out of 10.00

Assume that the kernel has assigned the linear address space between 0x30000000 and 0x30FFFFFF to a running process. Two-level Paging is used with page size 4K.

Suppose that the process needs to read the byte at linear address 0x30F42646 then determine which entry in the Page Directory table is accessed.

(write your answer in hex)

Answer: 0xC

The correct answer is: 0x0C3

Question 95

Correct

Mark 5.00 out of 5.00

Consider four-level Paging unit using a 48-bit virtual address divided into five fields -   Global Page directory containing most significant 9 bits, Upper Page directory containing next 9 bits, Middle Page directory containing next 9 bits, Page Table containing next 9 bits and page offset containing least significant 12 bits.

Determine the total number of Global Page Directory Tables.

Select one:

1. 256
2. 1
3. 1024
4. 512

The correct answer is: 1

Question 96

Correct

Mark 10.00 out of 10.00

Consider two-level Paging unit using a 32-bit virtual address divided into three fields -Page directory containing most significant 10 bits, Page Table containing intermediate 10 bits and page offset containing least significant 12 bits. Determine the maximum number of pages that can be stored.

Select one:

1. 1024 x 1024
2. 2048 x 2048
3. 4096 x 4096
4. 512 x 512

The correct answer is: 1024 x 1024

Question 97
Correct
Mark 10.00 out of 10.00

In Intel processors, paging is enabled by setting the ……. Flag in the …….. Register.

Select one:
- 1. PAE, cr0
- 2. PG , cr4
- 3. PG, cr0 ✓
- 4. PAE, cr4

The correct answer is: PG, cr0

Question 98
Incorrect
Mark 0.00 out of 5.00

The value 0 in 2-bit Requestor Privilege Level(RPL) field of CS register refers to …….. mode.

Answer: Kernel mode ✗

The correct answer is: kernel

Question 99
Correct
Mark 5.00 out of 5.00

Consider two-level Paging unit using a 32-bit virtual address divided into three fields -Page directory containing most significant 10 bits, Page Table containing intermediate 10 bits and page offset containing least significant 12 bits.

Determine the maximum number of Page tables that can be stored

Select one:
- 1. 4096
- 2. 512
- 3. 2048
- 4. 1024 ✓

The correct answer is: 1024

Question 100
Correct
Mark 10.00 out of 10.00

Consider four-level Paging unit using a 48-bit virtual address divided into five fields - Global Page directory containing most significant 9 bits, Upper Page directory containing next 9 bits, Middle Page directory containing next 9 bits, Page Table containing next 9 bits and page offset containing least significant 12 bits.

Determine the maximum number of Page tables

Select one:
- 1. 2048x2048X2048
- 2. 512x512X512 ✓
- 3. 256x256X256
- 4. 1024x1024X1024

The correct answer is: 512x512X512

Question 101

Incorrect

Mark 0.00 out of 10.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x00000000, Limit = 0xFFFFF, G=1

Determine the ending address of the Linear Segment in hex

Answer: 0xFFFFFFFFF

The correct answer is: 0xFFFFFFFF

Question 102

Incorrect

Mark 0.00 out of 10.00

Extended paging coexists with regular paging and is enabled by setting the ………flag of the ………….. processor control register.

Select one:

1. PAE, cr0
2. PSE, cr0
3. PSE, cr4
4. PAE , cr4

The correct answer is: PSE, cr4

Question 103

Correct

Mark 5.00 out of 5.00

Size of each Segment Descriptor stored in GDT or LDT is ....

Select one:

1. 64 bytes
2. 16 bytes
3. 8 bytes
4. 32 bytes

The correct answer is: 8 bytes

Question 104

Correct

Mark 10.00 out of 10.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x20000000, Limit = 0xFFFFF, G=0

Compute the size or length of the Linear Segment in MB.

Answer: 1

The correct answer is: 1MB

Question 105
Correct
Mark 10.00 out of 10.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x03000000, Limit = 0x3FFF, G=1

Compute the size or length of the Linear Segment in MB

Answer: 64

The correct answer is: 64MB

Question 106
Correct
Mark 10.00 out of 10.00

Consider two-level Paging unit using a 32-bit virtual address divided into three fields -Page directory containing most significant 10 bits, Page Table containing intermediate 10 bits and page offset containing least significant 12 bits.

Determine the maximum number of pages that can be stored and total amount of main memory required to store all these pages

Select one:
1. 2048x2048, 16GB
2. 1024x1024, 4GB
3. 4096x4096, 64GB
4. 512x512, 1GB

The correct answer is: 1024x1024, 4GB

Question 107
Correct
Mark 5.00 out of 5.00

A page table entry contains ………………………………..

Select one:
1. base address of each page in physical memory
2. page size
3. page
4. page offset

The correct answer is: base address of each page in physical memory

Question 108
Correct
Mark 5.00 out of 5.00

The ....register in the processor provides the base address of the Global Descriptor Table in the main memory.

Select one:
1. CR3
2. gdtr
3. CR0
4. ldtr

The correct answer is: gdtr

Question 109
Correct
Mark 5.00 out of 5.00

Many kernels also implement a .... system call, which allows a process to wait for a specific child process to reap when it terminates.

Answer: waitpid()

The correct answer is: waitpid( )

Question 110
Correct
Mark 10.00 out of 10.00

Suppose the gdtr register in the processor has the value 0x00900000 and the upper 13 bits of the CS register is used as the Segment Selector and contains the value 0x0D. What is the address of the corresponding Segment Descriptor stored in the GDT (i.e, address of the corresponding GDT entry).

Select one:

1. 0x00900068
2. 0x00900000
3. 0x00900104
4. 0x0090000D

The correct answer is: 0x00900068

Question 111
Incorrect
Mark 0.00 out of 10.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x03000000, Limit = 0x3FFF, G=0

Determine the ending address of the Linear Segment in hex.

Answer: 0x03003FFF

The correct answer is: 0x3003FFF

Question 112
Correct
Mark 5.00 out of 5.00

Run time mapping from linear to physical address is done by ....

Select one:

1. I/O
2. PCI
3. CPU
4. MMU

The correct answer is: MMU

Question 113
Correct
Mark 5.00 out of 5.00

In multiprocessor systems, there is …….. GDT for every CPU in the system.

Select one:
1. 8
2. 1
3. 16
4. 32

The correct answer is: 1

Question 114
Correct
Mark 2.00 out of 2.00

……… . Generates an exact copy of the current process that differs from the parent process only in its PID.

Select one:
1. Fork
2. New
3. Exec
4. Init

The correct answer is: Fork

Question 115
Correct
Mark 10.00 out of 10.00

Assume that the kernel has assigned the linear address space segment between 0x30000000 and 0x300FFFFF to a running process.

Two-level Paging is used with page size 4K. Determine the number of pages allocated in the segment for the process.

Select one:
1. 1024
2. 2048
3. 512
4. 256

The correct answer is: 256

Question 116
Correct
Mark 5.00 out of 5.00

The size of a page is typically : ………………………….

Select one:
1. power of 16
2. power of 4
3. varied
4. power of 2

The correct answer is: power of 2

Question 117
Incorrect
Mark 0.00 out of 3.00

In order to extend the size of the virtual address space usable by the processes, the Unix operating system makes use of .... on disk.

Answer: Swap space ✗

The correct answer is: swap areas

Question 118
Correct
Mark 10.00 out of 10.00

KMA stands for ....

Answer: Kernel Memory Allocator ✓

The correct answer is: Kernel Memory Allocator

Question 119
Correct
Mark 10.00 out of 10.00

In 80x86 processors, Logical address associated with stack is specified using the registers ....

Select one:
1. DS, SI
2. DS, DI
3. CS, IP
4. SS, SP ✓

The correct answer is: SS, SP

Question 120
Correct
Mark 10.00 out of 10.00

A segment descriptor entry in GDT contains the following fields :

Base (starting address of linear segment)= 0x20000000, Limit = 0xFFFFF, G=1

Compute the size or length of the Linear Segment.

Answer: 4 GB ✓

The correct answer is: 4 GB

Question 121
Correct
Mark 5.00 out of 5.00

Process 0 is also known as .... process

Answer: swapper ✓

The correct answer is: swapper

Question 122

Correct

Mark 5.00 out of 5.00

……………………….. translates logical addresses to linear addresses

Select one:

1. Paging Unit
2. Segmentation Unit ✓
3. Arithmetic Logic Unit
4. Memory Controller unit

The correct answer is: Segmentation Unit

Question 123

Correct

Mark 10.00 out of 10.00

In 80x86 processors, Logical address associated with data is specified using the registers ………………………

Select one:

1. CS, IP
2. CS, DI
3. SS, SP
4. DS, SI ✓

The correct answer is: DS, SI

Question 124

Correct

Mark 20.00 out of 20.00

Consider four-level Paging unit using a 48-bit virtual address divided into five fields - Global Page directory containing most significant 9 bits, Upper Page directory containing next 9 bits, Middle Page directory containing next 9 bits, Page Table containing next 9 bits and page offset containing least significant 12 bits.

Determine the maximum amount of memory required to store Page directory tables, Page tables & all pages, assuming that each entry in the page directory tables and page tables take 64-bits (each page table entry size =64bits).

Select one:

1. 8KB+8MB+8GB+8TB+4PB
2. 2KB+512KB+128MB+32GB+16TB
3. 4KB+2MB+1GB+512GB+256TB ✓
4. 16KB+32MB+64GB+128TB+64PB

The correct answer is: 4KB+2MB+1GB+512GB+256TB

Question 125
Incorrect
Mark 0.00 out of 8.00

Assume that the kernel has assigned the linear address space between 0x30000000 and 0x30FFFFFF to a running process. Two-level Paging is used with page size 4K.

Suppose that the process needs to read the byte at linear address 0x30F42646 then determine which entry in the Page Table is accessed.

Answer: 0x3D0 ✗

The correct answer is: 0x342

Question 126
Incorrect
Mark 0.00 out of 5.00

A .... is very similar to a semaphore, but it has no process list: when a process finds the lock closed by another process, it revolves around it repeatedly, executing a tight instruction loop until the lock becomes open.

Answer: Spinlock ✗

The correct answer is: spin lock

Question 127
Correct
Mark 5.00 out of 5.00

The segment limit contains the:

Select one:
- 1. starting physical address of the segment in memory
- 2. starting logical address of the process
- 3. starting linear address of the process
- 4. segment length ✓

The correct answer is: segment length

Question 128
Correct
Mark 5.00 out of 5.00

The offset of the logical address must be :

Select one:
- 1. between 0 and the segment number
- 2. greater than the segment number
- 3. between 0 and segment limit ✓
- 4. greater than segment limit

The correct answer is: between 0 and segment limit

Question 129

Incorrect

Mark 0.00 out of 3.00

Is the following statement TRUE or FALSE ………………….

"In uniprocessor systems there is only one GDT, while in multiprocessor systems there is one GDT for every CPU in the system".

Select one:

- True
- False ❌

The correct answer is 'True'.